Shashi Bhusan Mishra

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RESEARCH SUMMARY

I am a theoretical and computational condensed matter physicist developing predictive first-principles approaches to understand and design quantum materials. My research connects crystal and magnetic symmetry with many-body interactions to uncover the microscopic mechanisms governing superconductivity, unconventional magnetism, topological transport, and light-driven phenomena. I combine density-functional and many-body theories with Wannier-based methods, spin modeling, Berry-phase approaches, and high-performance computing, while developing open-source electron–phonon and superconductivity capabilities in EPW and EPWpy. My independent research program will build on this framework to predict and control emergent electronic, magnetic, and superconducting phases in quantum materials, including their response to symmetry breaking, interfaces, and external fields.

ACADEMIC APPOINTMENTS

Assistant Research Scientist March 2026 – present
Department of Electrical and Computer Engineering, University of Maryland, College Park, USA
Group Head: Prof. Thomas E. Murphy

Postdoctoral Fellow April 2023 – March 2026
Department of Physics, SUNY Binghamton, NY, USA
Advisor: Prof. Roxana Margine

Postdoctoral Fellow April 2021 – March 2023
Department of Mechanical Engineering, University of California, Riverside, USA
Advisor: Prof. Sinisa Coh

EDUCATION

Ph.D. in Physics March 2021
Indian Institute of Technology Madras, India
Thesis: Ab initio Modelling and Functionalization of Graphene & TiO_2 Surfaces and Interfaces
Advisors: Prof. B. R. K. Nanda & Prof. Somnath C. Roy

M.Sc. in Physics June 2013
Utkal University, Odisha, India

B.Sc. in Physics June 2011
Fakir Mohan University, Odisha, India

RESEARCH INTERESTS

- Quantum magnetism and spintronics
- Superconductivity and electron–phonon physics
- Topological quantum materials
- Quantum transport and Berry-phase phenomena
- Light–matter interactions
- First-principles materials discovery

RESEARCH DIRECTIONS

- **Symmetry-guided quantum magnetism:** Combine magnetic symmetry, first-principles calculations, and spin models to predict altermagnets, non-collinear and chiral antiferromagnets, and two-dimensional magnets, and to connect microscopic exchange interactions to magnetic ordering, spin textures, and transverse transport.
- **Interfacial control of magnetism and multiferroicity:** Design van der Waals and oxide heterostructures in which ferroelectric polarization and interfacial symmetry breaking tune charge transfer, magnetic anisotropy, and spin transport, enabling electrically controlled magnetic states.
- **Predictive theory of superconductivity:** Develop first-principles approaches beyond conventional Migdal–Eliashberg theory by incorporating anharmonicity, vertex corrections, full-bandwidth effects, and *ab initio* Coulomb interactions, with applications to hydrides, low-dimensional materials, and noncentrosymmetric and topological superconductors.
- **Topology and quantum transport:** Connect Berry-phase physics with first-principles transport to predict anomalous and spin Hall effects, magnetotransport, and magnetoelectric responses in Weyl, Dirac, and magnetic topological materials.

- **Light-driven quantum materials:** Develop first-principles descriptions of ultrafast spin and orbital dynamics, including optically induced magnetism and the inverse Faraday effect, to explore light-controlled magnetic and topological phases.
- **Data-driven materials discovery:** Integrate materials databases and machine learning with first-principles calculations to identify promising magnets and superconductors and prioritize candidates for detailed many-body modeling and experimental investigation.

PUBLICATIONS

30 peer-reviewed articles | 17 first/co-first-author papers | 11 manuscripts under review | 498 citations | h-index 15. * Equal contribution; author name in **bold**.

1. M. Keivanloo, **S. B. Mishra**, A. Shariatmanesh, M. Sandoghchi, K. Hongo, R. Maezono, E. R. Margine, S. Esmaeili, and M. Khazaei, "Superconductivity and band topology in functionalized 2D hexagonal MBenes," *Journal of Materials Chemistry C* (2026).
2. M. Keivanloo, F. Dinmohammad, **S. B. Mishra**, M. Sandoghchi, M. J. Arshia, M. Kawamura, E. R. Margine, M. H. Mahyuddin, H. Raebiger, R. P. P. Sukanli, K. Hongo, R. Maezono, and M. Khazaei, "High-T_c superconductivity in functionalized out-of-plane ordered double transition metal MXenes," *npj Computational Materials* (2026).
3. **S. B. Mishra***, Z. Liu*, S. Tiwari, F. Giustino, and E. R. Margine, "Comparative study of phonon-limited carrier transport in the Weyl semimetal TaAs family," *Physical Review B* **113**, 045202 (2026).
4. **S. B. Mishra** and R. Margine, "Nonadiabatic and anharmonic effects in high-pressure H_3S and D_3S superconductors," *Annalen der Physik* **538**, e00553 (2026).
5. S. Dey*, **S. B. Mishra***, A. Hazra, D. Kabiraj, and S. C. Roy, "Anomalous gas sensing characteristics of TiO_2 nanorod arrays irradiated with high-energy ion beam," *Materials Science and Engineering: B* **326**, 119210 (2026).
6. **S. B. Mishra**, H. Mori, and E. R. Margine, "Electron-phonon vertex correction effect in superconducting H_3S," *npj Computational Materials* **11**, 342 (2025).
7. **S. B. Mishra**, "Inverse Faraday effect in $3d$, $4d$, and $5d$ transition metals," *Physical Review B* **111**, 174413 (2025).
8. Z. Liu*, **S. B. Mishra***, J.-M. Lihm, S. Poncé, and E. R. Margine, "Phonon-limited carrier transport in the Weyl semimetal TaAs," *Physical Review B* **112**, 104311 (2025).
9. Y. Rambabu, **S. B. Mishra**, O. Henrotte, and E. Cortés, "Defects dynamics in photo-excited CeO_2 and their influence on CO_2 photoreduction," *Advanced Functional Materials*, e13933 (2025).
10. X. Zhang, **S. B. Mishra**, E. R. Margine, and E. Kioupakis, "Cubic BeB_2: a metastable p-type conductive material from first principles," *Physical Review B* **112**, 155206 (2025).
11. A. Das, S. Gumber, N. C. Maji, **S. B. Mishra**, Preethi M., P. Ning, R. G. Nair, Abhijith T, A. S. Vasenko, Madhumitha R., and O. V. Prezhdo, "Electrostatic self-assembly driven heterojunction of CeO_2/g-C_3N_4 nanosheets for efficient photocatalytic hydrogen evolution," *ACS Applied Materials & Interfaces* (2025).
12. **S. B. Mishra**, E. T. Marcial, S. Debata, A. N. Kolmogorov, and E. R. Margine, "Stability–superconductivity map for compressed Na-intercalated graphite," *Physical Review B* **110**, 174508 (2024).
13. P. Sahoo, **S. B. Mishra**, S. Debata, A. Sharma, B. Sahu, S. Padhan, R. Thangavel, and B.-K. Lee, "Hydrothermally grown halogen-doped ZnO nanorods for photoelectrochemical water oxidation: experimental and DFT insights," *Journal of Physical Chemistry C* **128**, 18711 (2024).
14. Abhijitha V. G., **S. B. Mishra**, and B. R. K. Nanda, "Si_2BN nanosheets as anchoring cathode material for realizing high capacity Al-ion battery," *Journal of Energy Storage* **77**, 109913 (2024).
15. V. Ortiz, **S. B. Mishra**, L. Vuong, S. Coh, and R. B. Wilson, "Transient ellipsometry measurements of the specular inverse Faraday effect in transition metals," *Physical Review Materials* **7**, 125202 (2023).
16. **S. B. Mishra** and S. Coh, "Spin contribution to the inverse Faraday effect of nonmagnetic metals," *Physical Review B* **107**, 214432 (2023).
17. B. Biswal, **S. B. Mishra**, R. Yadav, S. Poudyal, R. Rajarapu, P. K. Barman, K. R. Pandurang, M. Mandal, R. Singh, B. R. K. Nanda, and A. Misra, "Work function of van der Waals topological semimetals," *Applied Physics Letters* **120**, 093101 (2022).
18. Abhijitha V. G., **S. B. Mishra**, S. Ramaprabhu, and B. R. K. Nanda, "Design of aluminium ion battery with graphyne host," *Nanoscale Advances* **4**, 3870 (2022).

19. S. Dey, A. Chakravorty, **S. B. Mishra**, N. Khatun, A. Hazra, B. R. K. Nanda, S. Chandran, D. Kabiraj, and S. C. Roy, "High energy ion induced modification in TiO_2 nanorods," *Nanoscale Advances* **4**, 241 (2022).

20. **S. B. Mishra**, Abhijitha V. G., S. Ramaprabhu, and B. R. K. Nanda, "Graphdiyne—a promising cathode material for aluminium dual ion battery," *ACS Applied Energy Materials* **4**, 7786 (2021).

21. **S. B. Mishra**, S. Marutheeswaran, S. C. Roy, V. Natarajan, P. K. Rai, and B. R. K. Nanda, "Adsorption and degradation mechanism of 2,4,6-TNT on TiO_2(110) surface," *Surface Science* **713**, 171902 (2021).

22. J. Khatua, T. Arh, **S. B. Mishra**, H. Luetkens, A. Zorko, B. Sana, M. S. R. Rao, B. R. K. Nanda, and P. Khuntia, "Development of short- and long-range magnetic order in double perovskite Ba_2MnTeO_6," *Scientific Reports* **11**, 6959 (2021).

23. **S. B. Mishra**, S. C. Roy, and B. R. K. Nanda, "Electronic structure of graphene/TiO_2 interface: design and functional perspectives," *Applied Surface Science* **542**, 148709 (2021).

24. **S. B. Mishra**, S. K. Yadav, D. G. Kanhere, and B. R. K. Nanda, "Fluorine-intercalated graphene: formation of a two-dimensional spin lattice through pseudoatomization," *Physical Review Materials* **4**, 074411 (2020).

25. **S. B. Mishra** and B. R. K. Nanda, "Facet-dependent catalytic activities of anatase TiO_2 for CO_2 adsorption and conversion," *Applied Surface Science* **531**, 147330 (2020).

26. P. Panigrahi*, **S. B. Mishra***, T. Hussain, B. R. K. Nanda, and R. Ahuja, "Density functional theory studies of Si_2BN nanosheets as anode materials for magnesium ion batteries," *ACS Applied Nano Materials* **3**, 9055 (2020).

27. S. Marutheeswaran*, **S. B. Mishra***, S. C. Roy, and B. R. K. Nanda, "Mechanistic understanding of NO_2 dissociation on a rutile TiO_2(110) surface: an electronic structure study," *Journal of Physical Chemistry C* **124**, 8786 (2020).

28. **S. B. Mishra**, A. Choudhary, S. C. Roy, and B. R. K. Nanda, "Quantum-mechanical process of carbonate complex formation and large-scale anisotropy in the adsorption energy of CO_2 on anatase TiO_2(001) surface," *Physical Review Materials* **2**, 115801 (2018).

29. S. Samanta, **S. B. Mishra**, and B. R. K. Nanda, "Quantum well structure of double perovskite superlattice and formation of spin-polarized two-dimensional electron gas," *Physical Review B* **98**, 115155 (2018).

30. **S. B. Mishra** and B. R. K. Nanda, "Virtual synthesis of crystals using *ab initio* MD: case study on $LiFePO_4$," *AIP Conference Proceedings* **1832**, 090044 (2017).

Manuscripts under review

1. **S. B. Mishra**, F. Belli, E. Zurek, and E. R. Margine, "Anharmonicity and nonadiabaticity in hydride superconductors," arXiv:2608.14949 (2026) [Under review, *Physical Review Letters*].

2. A. Rahman*, **S. B. Mishra***, S. Khan, K. Shah, X. Ma, X. Lin, T. Sobol, L. Zhang, W. Zhao, M. Zhang, and Z. Muhammad, "Symmetry-enforced topology and Van Hove saddle points in hexagonal TiAs," arXiv:2610.12925 (2026).

3. V. Sharma, **S. B. Mishra**, S. Debata, P. Dev, G. M. Stephen, M. Zhukovskyi, T. Orlova, S. P. Bennett, F. M. Able, M. E. Jamer, A. L. Friedman, B. A. Assaf, and D. Heiman, "Topological Hall effect in nanocrystalline FeGe films," [Under review, *Physical Review Materials*].

4. Z. Muhammad*, **S. B. Mishra***, Gayatri, G. Krasucki, W. Ali, K. Olkowska-Pucko, O. Iqbal, Z. U. Rehman, A. U. Rahman, M. R. Molas, L. Xiaoyang, and W. Zhao, "Chiral phonons and giant anisotropic photoresponse in quasi-1D van der Waals semiconductor $ZrSnS_3$," arXiv:2608.22700 (2026) [Revision, *Laser and Photonics Reviews*].

5. S. Tiwari, J. Lafuente-Bartolomé, M. Zacharias, S. Jung, **S. B. Mishra**, T. Y. Kim, K. Yu, Z. Dai, A. Denchfield, J. Park, K. Luo, D. Kim, W. Yang, X. Zhang, B. Cucco, V.-A. Ha, W. G. Vandenberghe, S. Poncé, E. Kioupakis, E. R. Margine, and F. Giustino, "Programmable electron-phonon and many-body workflows with EPWpy," [Revision, *npj Computational Materials*].

6. S. Jois, **S. B. Mishra**, J. Dong, J. L. Spellberg, T. Xie, T. Alem, K. Burns, G. Gu, V. Kulichenko, L. Balicas, A. J. Sternbach, C. Gong, G. M. Stephen, P. Dev, A. T. Hanbicki, and A. L. Friedman, "Insulator-like behavior in $Bi_2Sr_2CaCu_2O_{8+x}$ driven by Fe_5GeTe_2 at room temperature," [Submitted to Science].

7. **S. B. Mishra**, S. ElMeligy, and P. Dev, "Superconductivity in noncentrosymmetric NbReSi: beyond harmonic and adiabatic limits," arXiv:2609.30564 (2026) [Submitted to *Physical Review B*].

8. **S. B. Mishra**, "Energetic near-degeneracy and anharmonic stabilization of the orthorhombic $Ima2$ phase of TaReSi," [Submitted to *Physical Review B*].

9. **S. B. Mishra**, "Comment on 'Comment on "Fully gapped superconductivity and topological aspects of the noncentrosymmetric superconductor TaReSi"'" [Submitted to *Physical Review B*].

10. I. Naumov*, **S. B. Mishra***, S. Shirodkar, S. Sheoran, and P. Dev, "Interplay between symmetry, topology and superconductivity in 2M-WS_2" [Submitted to *Physical Review X*].
11. N. W. G. Smith, **S. B. Mishra**, S. Jois, P. Dev, G. M. Stephen, S. Krylyuk, A. Davydov, L. Balicas, A. T. Hanbicki, and A. L. Friedman, "Magnetic-field-assisted resistive switching in α-In_2Se_3/Fe_5GeTe_2 heterostructures." [Submitted to *Nano Letters*].

Manuscripts in preparation

1. **S. B. Mishra**, R. Kumar, J. Paglione, and P. Dev, "Interlayer-frustrated RKKY magnetism in Gd_2AlSi_3: an *ab initio* spin model for the Néel transition, and a route to nanometre skyrmions in the R_2AlSi_3 family."
2. **S. B. Mishra** and P. Dev, "Raising the magnetic ordering scale of B20 FeGe: surfaces, lattice distortion, and local chemistry from first principles."
3. **S. B. Mishra**, S. Jois, and P. Dev, "Charge partitioning between the BiO reservoir and the CuO_2 plane in $Bi_2Sr_2CaCu_2O_{8+x}$: a pseudo-Jahn–Teller origin for the parent insulator and its stability against doping and a ferromagnetic contact."
4. **S. B. Mishra**, I. Naumov, and P. Dev, "Strain induced superconductivity dome in 2M-WS_2."
5. **S. B. Mishra**, R. Margine, and R. Akashi, "Role of higher-order van Hove singularity effect on self-energy of H_3S."
6. C. Dawra, **S. B. Mishra**, Z. Liu, and E. R. Margine, "Comparative study of phonon-limited carrier transport in the nodal-line semimetal ZrSiX (X = S, Se, Te) family."

RESEARCH GRANTS

- **Principal Investigator**, ACCESS Allocation PHY250052 (2025–2026): awarded 150,000 NSF-supported service-unit hours through national competitive review for first-principles simulations of quantum materials.

HONORS AND AWARDS

- APS DCOMP Travel Award (2026), American Physical Society Division of Computational Physics, Global Physics Summit, Denver, Colorado.
- Young Scientist Award (2024), Odisha Physical Society, India.
- Best Ph.D. Thesis Award in Physics (2021), Prof. A. L. Lashkar Prize, IIT Madras, India.
- DST Young Scientist Travel Award (2019), Department of Science and Technology, Govt. of India, for ICAMM, Rennes, France.
- Oral Presentation Award, EESTER International Conference, SRM and IIT Madras (2020), India.
- Best Poster Award, International Conference on Advanced Functional Materials (ICAFM), Anna University (2017), India.
- First Prize, I2CAM School on Clean and Renewable Energy, JNCASR, Bangalore (2017), India.
- Meritorious Master Scholarship, Institute of Mathematics & Applications, DST India (2011–2013).

TEACHING EXPERIENCE

Guest Lecturer, Statistical Mechanics I (PHYS 631), graduate course, Department of Physics, Binghamton University Fall 2025

Teaching Assistant, Department of Physics, IIT Madras 2015–2019
Assisted undergraduate and graduate laboratory and lecture courses, including Physics I/II, Engineering Physics Laboratory, M.Sc. Physics Laboratory, and Advanced Physics Laboratory.

Workshop and Short-Course Instructor

- Instructor, APS Short Course: *Electron-Phonon Physics from First Principles: Theory and Computational Tools*, APS Global Physics Summit, Denver, CO (2026).
- Instructor, EPW–Quantum ESPRESSO Summer School on Electron-Phonon Physics, UT Austin, TX (2024).
- Instructor, *EESTER–2018 Workshop* on Quantum ESPRESSO and first-principles methods, jointly organized by SRM IST and IIT Madras, India (2018).

Teaching Interests

- **Core M.Sc./graduate courses:** Quantum Mechanics I & II, Statistical Mechanics, Classical Mechanics, Electrodynamics, Condensed Matter Physics, Many-Body Theory.
- **B.Tech./undergraduate courses:** Introductory Physics (mechanics, waves, electromagnetism, optics, quantum) and laboratory, Solid State Physics, Mathematical Methods, Computational Physics.
- **Specialized electives:** Magnetism and Quantum Materials (exchange, anisotropy, magnetic symmetry, altermagnets, multiferroics, spin transport), First-Principles Electronic Structure Methods, Electron-Phonon Physics, Scientific Computing for Materials Physics.

MENTORING

- **Ph.D. students co-mentored (current):** Pawan Joshi (SUNY Binghamton), Sohair ElMeligy (Virginia Tech), and Benjamin Edun (University of Maryland), on projects in hydride superconductivity, phonon-mediated superconductivity, and silicon defects and heterointerfaces.
- **Ph.D. students co-mentored (previous):** Abhijitha V. G. (IIT Madras), on materials design for Al-ion batteries, resulting in four co-authored peer-reviewed publications in *J. Energy Storage*, *ACS Appl. Energy Mater.*, *Nanoscale Adv.*, and *ACS Appl. Nano Mater.*
- **Integrated M.Sc. students projects (IIT Madras):** Harshvardhan Murali (B.Tech. 2013–2017), on vacancy-induced magnetism in graphene; Sajid Ali (B.Tech. 2011–2016), on long-range magnetic ordering in graphene through halogen intercalation.

SOFTWARE AND WORKFLOW DEVELOPMENT

- Contribute to EPW and EPWpy, developing automated workflows for electron–phonon interactions, isotropic and anisotropic Migdal–Eliashberg superconductivity, and post-processing.
- Developed first-principles magnetism workflows connecting DFT and magnetic force-theorem calculations to exchange interactions, magnetic anisotropy, and Monte Carlo/mean-field ordering temperatures, including machine-learning-assisted generation of amorphous magnetic structures.
- Implemented Weyl-node detection, adaptive Berry-curvature sampling, Berry-curvature corrections to Boltzmann transport, and two-level MPI parallelization of the EPW superconductivity module; derived and implemented the inverse Faraday effect for metals in WannierBerri.
- Contribute to Superconductivity and EPW documentation, including hands-on tutorials for electron–phonon and superconductivity calculations.

PROFESSIONAL SERVICE

Leadership in Professional Societies

- Secretary/Treasurer, APS Topical Group on Energy Research and Applications (GERA) (2025–2027).
- Early Career Representative, APS Division of Computational Physics (DCOMP) (2024–2025).
- APS Annual Leadership Meeting and Congressional Visit: Representative for DCOMP, Washington, DC (January 2025).

Conference and Session Organization

- Session Chair, APS Global Physics Summit 2026, Session S46: "Exciton–Phonon Interactions", Denver, CO (2026).
- Session Chair, APS March Meeting 2024, Session K58: "Electrons, Phonons, and Electron-Phonon Scattering".
- Participant and presenter, NSF Cyberinfrastructure for Sustained Scientific Innovation (CSSI) PI Meeting, Denver, CO (July 2025).

Peer Review

- Reviewed 30 manuscripts for journals including *Physical Review Letters*, *Physical Review Applied*, *Scientific Reports*, *Computer Physics Communications*, *Journal of Energy Storage*, *International Journal of Hydrogen Energy*, and *Journal of Alloys and Compounds*.

Professional Memberships

- Member, Sigma Xi: The Scientific Research Honor Society (University of Maryland Chapter, 2026–present).
- Member, American Physical Society (APS), including the Divisions of Computational Physics (DCOMP), Condensed Matter Physics (DCMP), and Topical Group on Energy Research and Applications (GERA) (2021–present).

CONFERENCE PRESENTATIONS

Invited Talks

- "Insights into the surface chemistry of oxide semiconductors," *ICAM-SEEi 2025 Workshop*, IITRAM, Ahmedabad, India (November 27, 2025).
- "Understanding phonon-mediated superconductivity: from fundamentals to frontiers," *International Faculty Development Program on Frontiers in Material Science*, SRM IST, Chennai, India (March 21, 2025).
- "Phonon-mediated superconductivity in H_3S and NaC_4," *Young Scientist Award Winner Talk, 41st Annual Convention of Orissa Physical Society*, KIIT University, Bhubaneswar, India (February 8, 2025).
- "An introduction to electronic properties of materials and magnetism in graphene," *Department of Physics Seminar, Fakir Mohan Autonomous College*, Balasore, India (January 25, 2019).

Selected Contributed Talks (from 30 total)

- "Anharmonic and nonadiabatic electron-phonon effects in hydride superconductors," *APS Global Physics Summit*, Denver (2026, accepted).
- "Automating many-body workflows with EPWpy in the MATCSSI framework," *NSF CSSI-PI Meeting*, Denver (2025).
- "Stability–superconductivity map for compressed Na-intercalated graphite," *APS March Meeting*, Anaheim (2025).
- "Phonon-mediated superconductivity beyond the Migdal approximation," *APS March Meeting*, Minneapolis (2024).
- "Theory of inverse Faraday effect in $3d$, $4d$, and $5d$ metals," *APS March Meeting*, Las Vegas (2023).

COMPUTATIONAL EXPERTISE

- **Electronic structure & magnetism:** DFT, DFPT, DFT+U, spin–orbit coupling, noncollinear magnetism, magnetic symmetry, exchange interactions and magnetic anisotropy, spin models, and Monte Carlo methods.
- **Superconductivity & lattice dynamics:** Electron–phonon coupling, isotropic and anisotropic Eliashberg theory, vertex corrections, anharmonic phonons, *ab initio* molecular dynamics, and machine-learned force fields.
- **Topology & transport:** Wannier interpolation, Berry-phase methods, topological invariants, Boltzmann transport, and anomalous and spin Hall responses.
- **Codes:** Quantum ESPRESSO, EPW/EPWpy, VASP, ABINIT, Wannier90, WannierBerri, WannierTools, and IRVSP.
- **Programming & HPC:** Fortran, Python, Bash, Git, MPI/OpenMP, SLURM, and large-scale high-performance computing.